CSE 320 - Computer Networks

LAB Session B

09.03.2026

Subnetting and Static Routing Between Multiple Networks

A. Further Network Exercise

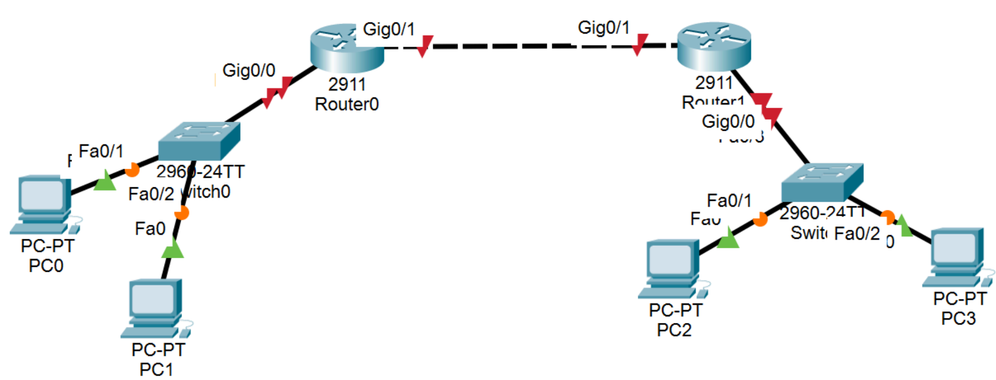
In this lab, you will learn how to:

- divide a Class C network into smaller subnetworks
- assign those subnetworks to router and switch interfaces
- configure end devices with proper IP addresses
- connect two routers using a separate transit network
- add **static routes** so different subnetworks can communicate

This lab moves from simple VLAN work into actual **Layer 3 network design**.

1. Topology

Careful about the interface plan :)



2. Addressing task

Base Network:

10.10.40.0/24

You will divide it into **four equal /27 subnets**.

Each /27 subnet has:

- **32 total addresses**
- **30 usable host addresses**

Subnet Mask:

255.255.255.224

3. Subnet names and assignment

Subnet Name	Network Address	Usable Host Range	Broadcast	Purpose
Admin LAN	10.10.40.0/27	10.10.40.1 – 10.10.40.30	10.10.40.31	PC0 / VLAN 10
Research LAN	10.10.40.32/27	10.10.40.33 – 10.10.40.62	10.10.40.63	PC1 / VLAN 20
Services LAN	10.10.40.64/27	10.10.40.65 – 10.10.40.94	10.10.40.95	PC2 / VLAN 30
Backup LAN	10.10.40.96/27	10.10.40.97 – 10.10.40.126	10.10.40.127	PC3 / VLAN 40
Transit Network	10.10.40.128/27	10.10.40.129 – 10.10.40.158	10.10.40.159	Router-to-router link

4. Full addressing table

Router interfaces

Subnet Name	Network Address	Usable Host Range	Broadcast
Admin LAN	10.10.40.0/27	10.10.40.1 – 10.10.40.30	10.10.40.31
Research LAN	10.10.40.32/27	10.10.40.33 – 10.10.40.62	10.10.40.63
Services LAN	10.10.40.64/27	10.10.40.65 – 10.10.40.94	10.10.40.95
Backup LAN	10.10.40.96/27	10.10.40.97 – 10.10.40.126	10.10.40.127
Transit Network	10.10.40.128/27	10.10.40.129 – 10.10.40.158	10.10.40.159

PC addresses

Device	IP Address	Mask	Default Gateway
PC0	10.10.40.10	255.255.255.224	10.10.40.1
PC1	10.10.40.40	255.255.255.224	10.10.40.33
PC2	10.10.40.70	255.255.255.224	10.10.40.65
PC3	10.10.40.100	255.255.255.224	10.10.40.97

Cabling guide

Make the following connections:

Left side

- PC0 → SW1 Fa0/1
- PC1 → SW1 Fa0/2
- SW1 Fa0/24 → R1 G0/0
- R1 G0/1 can connect directly to another small switch/PC if you want a physically separate Student LAN

Right side

- PC2 → SW2 Fa0/1
- PC3 → SW2 Fa0/2
- SW2 Fa0/24 → R2 G0/0

Router-to-router link

- R1 G0/2 → R2 G0/1

5. VLAN Design

Switch0 VLANs:

10 -> Admin LAN, 20 -> Research LAN

Switch1 VLANs:

30 -> Services LAN, 40 -> Backup LAN

6. Configure Switch0

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW0
SW0(config)#
SW0(config)#vlan 10
SW0(config-vlan)#name Admin
SW0(config-vlan)#exit
SW0(config)#
SW0(config)#vlan 20
SW0(config-vlan)#name Research
SW0(config-vlan)#exit
SW0(config)#
SW0(config)#interface fa0/1
SW0(config-if)#switchport mode access
SW0(config-if)#switchport access vlan 10
SW0(config-if)#no shutdown
SW0(config-if)#exit
SW0(config)#
SW0(config)#interface fa0/2
SW0(config-if)#switchport mode access
SW0(config-if)#switchport access vlan 20
SW0(config-if)#no shutdown
SW0(config-if)#exit
SW0(config)#
SW0(config)#interface fa0/3
SW0(config-if)#switchport mode trunk

SW0(config-if)#no shutdown
SW0(config-if)#exit
SW0(config)#
SW0(config)#end
SW0#write memory
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Building configuration
```

7. Configure Switch1

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW1
SW1(config)#
SW1(config)#vlan 30
SW1(config-vlan)#name Services
SW1(config-vlan)#exit
SW1(config)#
SW1(config)#vlan 40
SW1(config-vlan)#name Backup
SW1(config-vlan)#exit
SW1(config)#
SW1(config)#interface fa0/1
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 30
SW1(config-if)#no shutdown
SW1(config-if)#exit
SW1(config)#
SW1(config)#interface fa0/2
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 40
SW1(config-if)#no shutdown
SW1(config-if)#exit
SW1(config)#
SW1(config)#interface fa0/3
SW1(config-if)#switchport mode trunk

SW1(config-if)#no shutdown
SW1(config-if)#exit
SW1(config)#
SW1(config)#end
SW1#write memory
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%SYS-5-CONFIG_I: Configured from console by console
```

8. Configure Router0 (Router-on-a-Stick)

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname RouterA
RouterA(config)#
RouterA(config)#interface g0/0
RouterA(config-if)#no shutdown

RouterA(config-if)#exit
RouterA(config)#
RouterA(config)#interface g0/0.10
RouterA(config-subif)#encapsulation dot1Q 10
RouterA(config-subif)#ip address 10.10.40.1 255.255.255.224
RouterA(config-subif)#exit
RouterA(config)#
RouterA(config)#interface g0/0.20
RouterA(config-subif)#encapsulation dot1Q 20
RouterA(config-subif)#ip address 10.10.40.33 255.255.255.224
RouterA(config-subif)#exit
RouterA(config)#
RouterA(config)#interface g0/1
RouterA(config-if)#ip address 10.10.40.129 255.255.255.224
RouterA(config-if)#no shutdown

RouterA(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
```

9. Configure Router1

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname RouterB
RouterB(config)#
RouterB(config)#interface g0/0
RouterB(config-if)#no shutdown

RouterB(config-if)#exit
RouterB(config)#
RouterB(config)#interface g0/0.30
RouterB(config-subif)#encapsulation dot1Q 30
RouterB(config-subif)#ip address 10.10.40.65 255.255.255.224
RouterB(config-subif)#exit
RouterB(config)#
RouterB(config)#interface g0/0.40
RouterB(config-subif)#encapsulation dot1Q 40
RouterB(config-subif)#ip address 10.10.40.97 255.255.255.224
RouterB(config-subif)#exit
RouterB(config)#
RouterB(config)#interface g0/1
RouterB(config-if)#ip address 10.10.40.130 255.255.255.224
RouterB(config-if)#no shutdown

RouterB(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0.40, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
```

10. Configure Static Routes

Router0

```
RouterA>enable
RouterA#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
RouterA(config)#ip route 10.10.40.64 255.255.255.224 10.10.40.130
RouterA(config)#ip route 10.10.40.96 255.255.255.224 10.10.40.130
RouterA(config)#end
RouterA#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
RouterA#
RouterA#
```

Router1

```
RouterB>
RouterB>
RouterB>enable
RouterB#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
RouterB(config)#ip route 10.10.40.0 255.255.255.224 10.10.40.129
RouterB(config)#ip route 10.10.40.32 255.255.255.224 10.10.40.129
RouterB(config)#end
RouterB#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
```

11. Verify

Check router interfaces

Check the following table with "`show ip interface brief`"

```
%SYS-5-CONFIG_I: Configured from console by console
write
Building configuration...
[OK]
RouterA#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  10.10.40.1      YES manual up                    up
GigabitEthernet0/0.20  10.10.40.33     YES manual up                    up
GigabitEthernet0/1     10.10.40.129    YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
RouterA#
```

```
RouterB#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.30  10.10.40.65     YES manual up                    up
GigabitEthernet0/0.40  10.10.40.97     YES manual up                    up
GigabitEthernet0/1     10.10.40.130    YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
RouterB#
```

Check routing table

"`show ip route`"

```
RouterA#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C       10.10.40.0/27 is directly connected, GigabitEthernet0/0.10
L       10.10.40.1/32 is directly connected, GigabitEthernet0/0.10
C       10.10.40.32/27 is directly connected, GigabitEthernet0/0.20
L       10.10.40.33/32 is directly connected, GigabitEthernet0/0.20
S       10.10.40.64/27 [1/0] via 10.10.40.130
S       10.10.40.96/27 [1/0] via 10.10.40.130
C       10.10.40.128/27 is directly connected, GigabitEthernet0/1
L       10.10.40.129/32 is directly connected, GigabitEthernet0/1

RouterA#
```

```
RouterB#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
S       10.10.40.0/27 [1/0] via 10.10.40.129
S       10.10.40.32/27 [1/0] via 10.10.40.129
C       10.10.40.64/27 is directly connected, GigabitEthernet0/0.30
L       10.10.40.65/32 is directly connected, GigabitEthernet0/0.30
C       10.10.40.96/27 is directly connected, GigabitEthernet0/0.40
L       10.10.40.97/32 is directly connected, GigabitEthernet0/0.40
C       10.10.40.128/27 is directly connected, GigabitEthernet0/1
L       10.10.40.130/32 is directly connected, GigabitEthernet0/1

RouterB#
```

12. Test Connectivity with ping

Don't forget to enter the PC ip address, subnet mask, and gateway address